

New Insight on Open-Structured Sodium Vanadium Oxide as High-Capacity and Long Life Cathode for Zn-Ion Storage: Structure, Electrochemistry, and First-Principles Calculation

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Herein, the promising properties of open-structured NaV_3O_8 as a cathode material for Zn-ion batteries (ZIBs) are investigated. First-principles calculations predict the insertion of Zn^{2+} (0.74 Å) in NaV_3O_8 with an interlayer distance of ≈ 7 Å, enabling delivery of a high discharge capacity of 353 mAh g⁻¹ at 70 mA g⁻¹ (0.2 C) for 300 cycles in the operating window of 0.3–1.5 V in 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ aqueous solution. Operando synchrotron X-ray diffraction, X-ray absorption near edge structure spectroscopy, and first-principles calculations validate the insertion of Zn^{2+} into the NaV_3O_8 structure within the operation range. Moreover, operando synchrotron X-ray diffraction and operando Raman spectroscopy reveal the formation of layered zinc hydroxytriflate ($\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$) as a side reaction below 0.8 V on discharge (reduction) and its dissolution into the electrolyte above 0.8 V on charge (oxidation). The formation of the Zn hydroxytriflate interfacial layer increases the charge-transfer activation energy from 15.5 to 48 kJ mol⁻¹, leading to kinetics fade below 0.8 V. The findings reveal the charge-storage mechanism for NaV_3O_8 , which may also be applicable to other vanadate cathodes, providing new insights for the investigation and design of ZIBs.

intensively investigated and developed at a global level. Among several possible energy storage and power sources, lithium-ion batteries (LIBs) have been successfully used in applications ranging from portable electronic devices to electric vehicles and large-scale energy storage systems (ESSs) owing to their high energy density. However, some critical concerns have arose regarding LIBs such as the cost of lithium resources and safety issues from mobile to ESS applications.^[1–6] Thus, rechargeable divalent-ion (e.g., Zn^{2+} , Mg^{2+} , and Ca^{2+}) batteries operated in aqueous electrolytes have recently received significant attention because the use of water results in low cost, improved safety, cell fabrication in ambient environment, and reasonable environmental impact.^[7–9] Furthermore, the ionic conductivity of aqueous electrolytes for rechargeable divalent-ion batteries is several-orders-of-magnitude higher than that of organic

electrolytes.^[7,10–12] Among divalent-ion batteries, aqueous zinc-ion batteries (ZIBs) have been intensively investigated owing to the high gravimetric capacity (820 mAh g⁻¹) and sufficient earth abundance of Zn metal. Moreover, the high overpotential of

1. Introduction

To mitigate recent environmental issues caused by fossil fuels, renewable energy conversion, and storage systems have been

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the hydrogen evolution reaction on Zn electrodes makes them stable in neutral or slightly acidic solutions and leads to a moderate potential window (≈ 2 V).^[7,10]

The first cathode material proposed for ZIBs was MnO_2 , for which Zn^{2+} intercalation/deintercalation is possible for repeated cycles.^[13] The electrochemical properties of different MnO_2 polymorphs such as α ,^[14] β ,^[15] γ ,^[16] and δ - MnO_2 ^[17] were shown to be related to complex mechanisms such as Zn^{2+} and H^+ co-intercalation or conversion rather than the typical intercalation of zinc ions, in which the electrochemical reduction accompanies parasitic deposition of $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot 5\text{H}_2\text{O}$ onto the active materials while the precipitate is diminished such that the active material is recovered to MnO_2 on oxidation.^[14,18] The disproportionation phenomenon of Mn^{3+} toward Mn^{2+} and Mn^{4+} is another concern that occurs in acidic media.

These phenomena have triggered the search for different types of cathode materials including Prussian blue analogues,^[19,20] Co_3O_4 ,^[21,22] $\text{VO}_2(\text{B})$,^[23,24] $\text{Zn}_3\text{V}_2\text{O}_7(\text{OH})_2$,^[25] VS_2 ,^[26] and Mo_6S_8 .^[27] Among these materials, vanadium oxides are of interest because, in addition to having various oxidation states from +2 to +5 for V_xO_y (e.g., VO , V_2O_3 , VO_2 , and V_2O_5), they also have structure motifs such as VOOH ,^[28] $\text{V}_3\text{O}_7 \cdot \text{H}_2\text{O}$,^[29] and $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$.^[30] These compounds are activated by the Zn^{2+} intercalation reaction. Layered $\text{V}_3\text{O}_7 \cdot \text{H}_2\text{O}$ ^[29] and $\text{Zn}_{0.25}\text{V}_2\text{O}_5 \cdot n\text{H}_2\text{O}$ ^[31] delivered high capacities of 375 and 282 mAh g⁻¹ at 1 C, respectively. We recently demonstrated the insertion ability of Zn^{2+} into the open-structured $\text{VO}_2(\text{B})/\text{rGO}$, which delivered a high capacity of 350 mAh g⁻¹ with excellent capacity retention.^[23] More progress has been made in alkali vanadate compounds such as AV_xO_y ($\text{A} = \text{Li}^+$, Na^+ , and K^+) owing to their open-frame structure.^[32–40] Alfuruqi et al.^[32] reported LiV_3O_8 that allows Zn^{2+} intercalation and delivers a high capacity of 275 mAh g⁻¹ (0.1 C). Mai et al.^[36] suggested the feasibility of Zn^{2+} insertion into $\text{Na}_2\text{V}_6\text{O}_{16} \cdot 1.63\text{H}_2\text{O}$ with a high specific capacity of 352 mAh g⁻¹ (50 mA g⁻¹). Zn^{2+} insertion was also demonstrated in $\text{K}_2\text{V}_6\text{O}_{16} \cdot 2.7\text{H}_2\text{O}$.^[38] It is evident that these alkali vanadates provide high capacity and acceptable cyclability; however, the associated reactions progressed via the intercalation of Zn^{2+} into the structure, which was always accompanied by an additional side reaction, forming a new phase when the cathode materials were discharged (reduced) below 0.8 V versus Zn^{2+}/Zn in ZnSO_4 aqueous solutions. Notably, the formed phase disappeared during charge (oxidation), and the process was highly reversible throughout cycling. He et al.^[40] named the new phase as $\text{Zn}_x\text{H}_2\text{V}_3\text{O}_8$ or $\text{Zn}_x\text{Na}_{0.33}\text{V}_2\text{O}_5$, whereas Guo et al.^[41] called the phase to be $\text{Zn}_4\text{V}_2\text{O}_9$. In contrast, Soundharajan et al.^[36] indexed the new phase as a zinc hydroxysulfate $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot 5\text{H}_2\text{O}$, which is a common deposit for transition-metal-oxide cathode materials for aqueous ZIBs.

These findings motivated us to investigate NaV_3O_8 not only to optimize the electrode performance but also mechanistically to identify the new phase that emerges at low voltage, evaluate kinetics of electrode process, and verify Zn-intercalation. Indeed, NaV_3O_8 is of interest in terms of structure because the interlayer distance, ≈ 7 Å, is sufficient for the insertion of Zn^{2+} (0.74 Å). This structural preference enables delivery of a high discharge capacity of 353 mAh g⁻¹ at 70 mA g⁻¹ (0.2 C) in the operating window of 0.3–1.5 V in 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ aqueous solution, which agrees with the result predicted by first-principle calculations, with excellent cyclability for long-term cycles.

Operando synchrotron X-ray diffraction (o-SXRD), X-ray absorption near edge structure (XANES) spectroscopy, and first-principle calculations validated the insertion of Zn^{2+} into the NaV_3O_8 structure within the operation range. O-SXRD and operando Raman spectroscopy proved that the formation of layered zinc hydroxytriflate ($\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$) always appears as a side reaction below 0.8 V on discharge, but this product dissolves into the electrolyte above 0.8 V on charge. Namely, along with Zn^{2+} intercalation, we identify highly reversible formation/dissolution of layered Zn hydroxyl triflate as an interfacial layer on the cathode upon discharge/charge. According to electrochemical impedance analysis, the formation of the Zn hydroxytriflate interfacial layer slows down the electrochemical kinetics, resulting in capacity fade at high discharge rates below 0.8 V. The obtained understanding of the NaV_3O_8 electrode reaction process provides insight both for Zn^{2+} storage and for the design of new types of electrode materials for ZIB.

2. Results and Discussion

The crystal structure and chemical composition of the as-synthesized product was analyzed using inductively coupled plasma-atomic emission spectroscopy (ICP-AES), with NaV_3O_8 identified, as designated. Therefore, Rietveld refinement of the X-ray diffraction (XRD) data was performed assuming the monoclinic structure with space group $P2_1/m$ (Figure 1a and Table S1, Supporting Information). The observed pattern coincided with the calculated one, with lattice parameters of $a = 7.281(3)$ Å, $b = 3.598(2)$ Å, $c = 12.049(2)$ Å, and $\beta = 107.12^\circ$. In addition, the V K-edge spectrum of NaV_3O_8 was almost identical to that of V_2O_5 , indicating that the average oxidation state of V was close to 5+ for NaV_3O_8 (Figure 1b). The particle morphology of the synthesized NaV_3O_8 was observed using field-emission scanning electron microscopy (FE-SEM) (Figure 1c) and high-resolution transmission electron microscopy (HR-TEM) (Figure 1d). NaV_3O_8 consisted of 1D bundle-type particles composed of stick stacks, with stick lengths ranging from 50 nm to 3 μm (Figure 1c,d). Energy dispersive X-ray spectroscopy (EDS) mapping images confirmed the uniform distribution of Na, V, and O elements in the particle. Furthermore, HR-TEM imaging revealed a well-defined lattice fringe region with (001) planes of NaV_3O_8 (Figure 1e). The crystal structure of NaV_3O_8 obtained from the Rietveld refinement results is shown in Figure 1a. The structure consists of distorted octahedra $[\text{VO}_6]$ and trigonal bipyramid $[\text{VO}_5]$ connected by shared edges. The sodium ions positioned in octahedral sites between the vanadate sheets and sodium atoms serve as pillars between the vanadium oxide layers. The pillar ions stabilize the host structure during insertion and extraction of cations such as Zn^{2+} . In addition, the large interlayer distance of the open-structured NaV_3O_8 (≈ 7 Å), which is larger than that of $\text{VO}_2(\text{B})$ ($d = 4.08$ Å) and V_2O_5 ($d = 4.56$ Å), may allow facile accommodation of Zn^{2+} ions (0.74 Å, with a coordination number of six).

As we present in Figure 1a (inset), the monoclinic NaV_3O_8 has empty channels in the structure. To demonstrate the reaction mechanism of NaV_3O_8 during Zn^{2+} de/intercalation, confirmation of the Zn^{2+} sites in the NaV_3O_8 structure was needed. Thus, to predict the possible sites in which Zn ions can be intercalated, we performed bond valence energy landscape (BVEL)

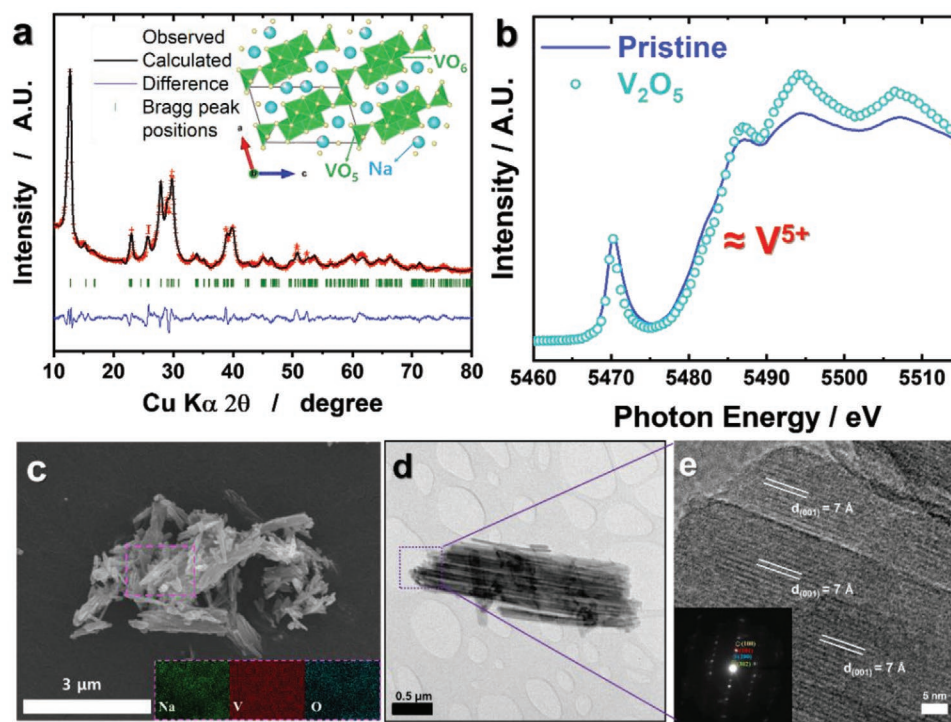


Figure 1. a) Rietveld refinement of XRD data for NaV_3O_8 (inset: crystal structural model of monoclinic $P2_1/m$); b) XANES spectra of pristine NaV_3O_8 for V K-edge; c) FE-SEM image of NaV_3O_8 ; d) HR-TEM images of NaV_3O_8 ; e) highlighted part of (d).

analyses using the *Bond_str* program. As shown in **Figure 2a**, it was predicted that large vacant sites are present between each layer composed of edge-shared distorted VO_6 octahedra along the a -axis direction, and a total of 2 mol Zn^{2+} ions per formula unit can be intercalated into the NaV_3O_8 structure. The predicted Wyckoff positions and atomic positions of possible Zn^{2+} sites are $2e$ (0.95, 0.25, 0.558), $2e$ (0.05, 0.25, 0.75), and $2e$ (0.8, 0.25, 0.35). Based on the structural information from the XRD and BVOL analyses with Rietveld refinement, we attempted to predict the theoretical properties of $\text{Zn}_x\text{NaV}_3\text{O}_8$ ($0 \leq x \leq 2$) using first-principles calculation. **Figure 2b** shows the predicted formation energies for various Zn^{2+} /vacancy configurations of $\text{Zn}_x\text{NaV}_3\text{O}_8$ compositions. To calculate the theoretical redox potentials of $\text{Zn}_x\text{NaV}_3\text{O}_8$ during Zn^{2+} de/intercalation, we used the formation energies and the following equation:

$$V = -\frac{E[\text{Zn}_{x_2}\text{NaV}_3\text{O}_8] - E[\text{Zn}_{x_1}\text{NaV}_3\text{O}_8] - (x_2 - x_1)E(\text{Zn})}{2(x_2 - x_1)F} \quad (1)$$

Here, V is the average redox potential between the compositions $\text{Zn}_{x_2}\text{NaV}_3\text{O}_8$ and $\text{Zn}_{x_1}\text{NaV}_3\text{O}_8$, and $E[\text{Zn}_x\text{NaV}_3\text{O}_8]$ is the lowest value among the formation energies of various Zn /vacancy configurations of $\text{Zn}_x\text{NaV}_3\text{O}_8$. It was verified that a total of 2 mol Zn^{2+} ions can be intercalated into the $\text{Zn}_x\text{NaV}_3\text{O}_8$ structure, which implies that the theoretical capacity of $\text{Zn}_x\text{NaV}_3\text{O}_8$ under the Zn-cell system is $\approx 353 \text{ mAh g}^{-1}$ in the voltage range between 0.3 and 1.5 V (vs Zn^{2+}/Zn). Moreover, the existence of several stable intermittent $\text{Zn}_x\text{NaV}_3\text{O}_8$ phases implies the single-phase reaction of $\text{Zn}_x\text{NaV}_3\text{O}_8$ during charge/discharge. **Figure 2c** shows that the theoretical redox potentials predicted by first-principles calculation. The intercalation

sequence of Zn^{2+} ions in $\text{Zn}_x\text{NaV}_3\text{O}_8$ predicted by first-principles calculation is shown in Table S2 (Supporting Information). For the most stable configuration of the $\text{Zn}_2\text{NaV}_3\text{O}_8$ composition, 3/4, 1, and 1/4 mol of Zn ions are in the Zn1, Zn2, and Zn3 sites, respectively. In addition, to confirm the structural change of $\text{Zn}_x\text{NaV}_3\text{O}_8$ during Zn^{2+} de/intercalation in detail, we arranged the crystal structures of the most stable $\text{Zn}_x\text{NaV}_3\text{O}_8$ compositions predicted by first-principles calculation (**Figure 3**). The total volume difference between $\text{Zn}_0\text{NaV}_3\text{O}_8$ and $\text{Zn}_2\text{NaV}_3\text{O}_8$ was verified to be less than $\approx 5\%$, despite intercalation of 2 mol Zn^{2+} ions into the structure. We supposed that the small volume change of $\text{Zn}_x\text{NaV}_3\text{O}_8$ during Zn^{2+} de/intercalation results from the transformation from pyramidal VO_5 to octahedral VO_6 with the displacement of V ions. The phase stability of $\text{Zn}_x\text{NaV}_3\text{O}_8$ is reduced with Zn^{2+} intercalation into the structure, which requires the particular local structural change to decrease the instability of $\text{Zn}_x\text{NaV}_3\text{O}_8$ phase. As shown in **Figure 3**, with transformation from pyramidal VO_5 to octahedral VO_6 during intercalation of 2 mol divalent Zn^{2+} ions, simple point-sharing between VO_5 and VO_6 is transformed to edge-sharing between VO_6 octahedra. This local structural change of $\text{Zn}_x\text{NaV}_3\text{O}_8$ can result in strengthened structural stability of $\text{Zn}_x\text{NaV}_3\text{O}_8$, enabling not only the small volume change of $\text{Zn}_x\text{NaV}_3\text{O}_8$ but also its outstanding cycle performance.

The synthesized NaV_3O_8 was tested in coin cells in the voltage range of 0.3–1.5 V at 0.2 C (70 mA g^{-1}) (**Figure 4a** and **Figure S1a**, Supporting Information). The discharge curves present a feature with two different slopes, namely, between 0.8 and 1.4 V and between 0.3 and 0.8 V. The $\text{NaV}_3\text{O}_8/\text{Zn}$ cell exhibited a high discharge capacity of $353 \text{ mAh (g-oxide)}^{-1}$ in

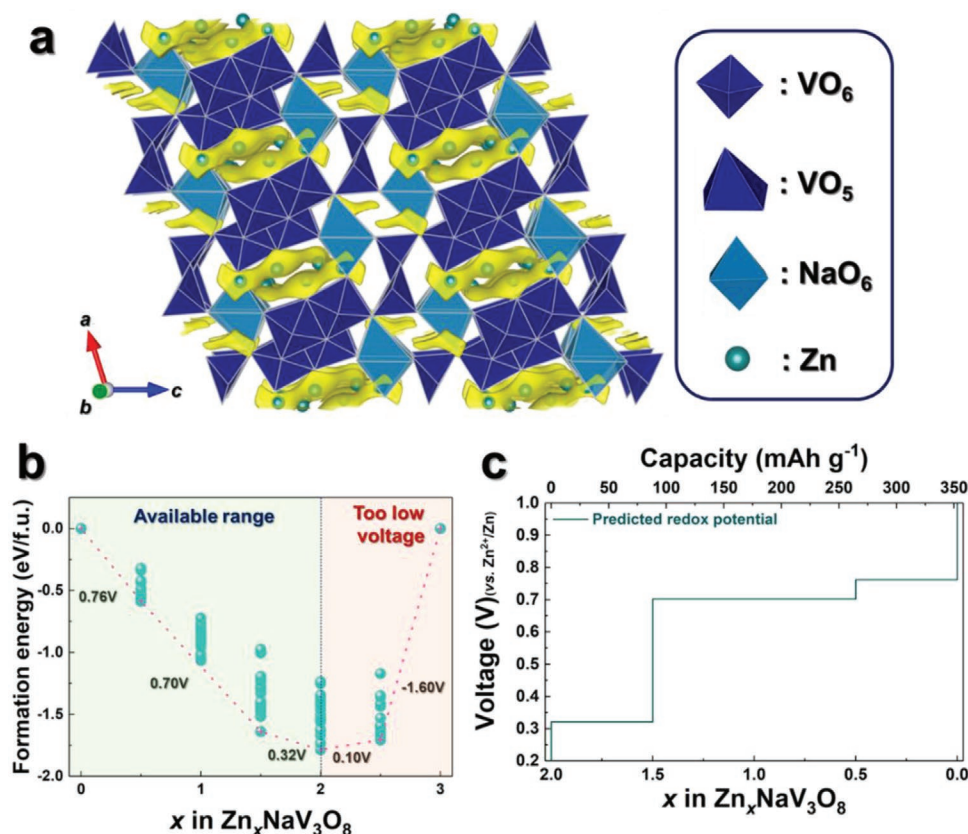


Figure 2. a) BVEL (bond valence energy landscape) analyses of $\text{Zn}_x\text{NaV}_3\text{O}_8$ with possible sites in which Zn^{2+} can be intercalated; b) convex-hull plot of theoretical formation energies of various $\text{Zn}_x\text{NaV}_3\text{O}_8$; and c) redox potentials of $\text{Zn}_x\text{NaV}_3\text{O}_8$ predicted by first-principles calculation.

the first cycle, and the charge capacity was sufficiently high (352.3 mAh g^{-1}), leading to a Coulombic efficiency (CE, charge capacity divided by discharge capacity) close to 100% (Figure 4a).

For the 300th cycle, the obtained capacity was $\approx 278 \text{ mAh g}^{-1}$, which represents 78% of the first capacity (Figure 4b and Figure S1a, Supporting Information). The $\text{NaV}_3\text{O}_8/\text{Zn}$ cell

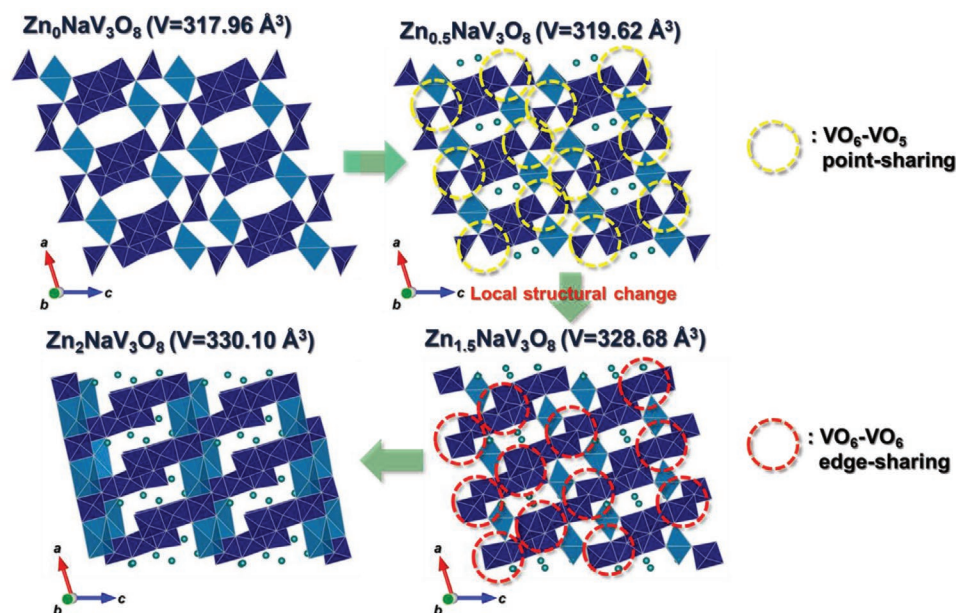


Figure 3. Predicted crystal structures of $\text{Zn}_x\text{NaV}_3\text{O}_8$ based on first-principles calculation.

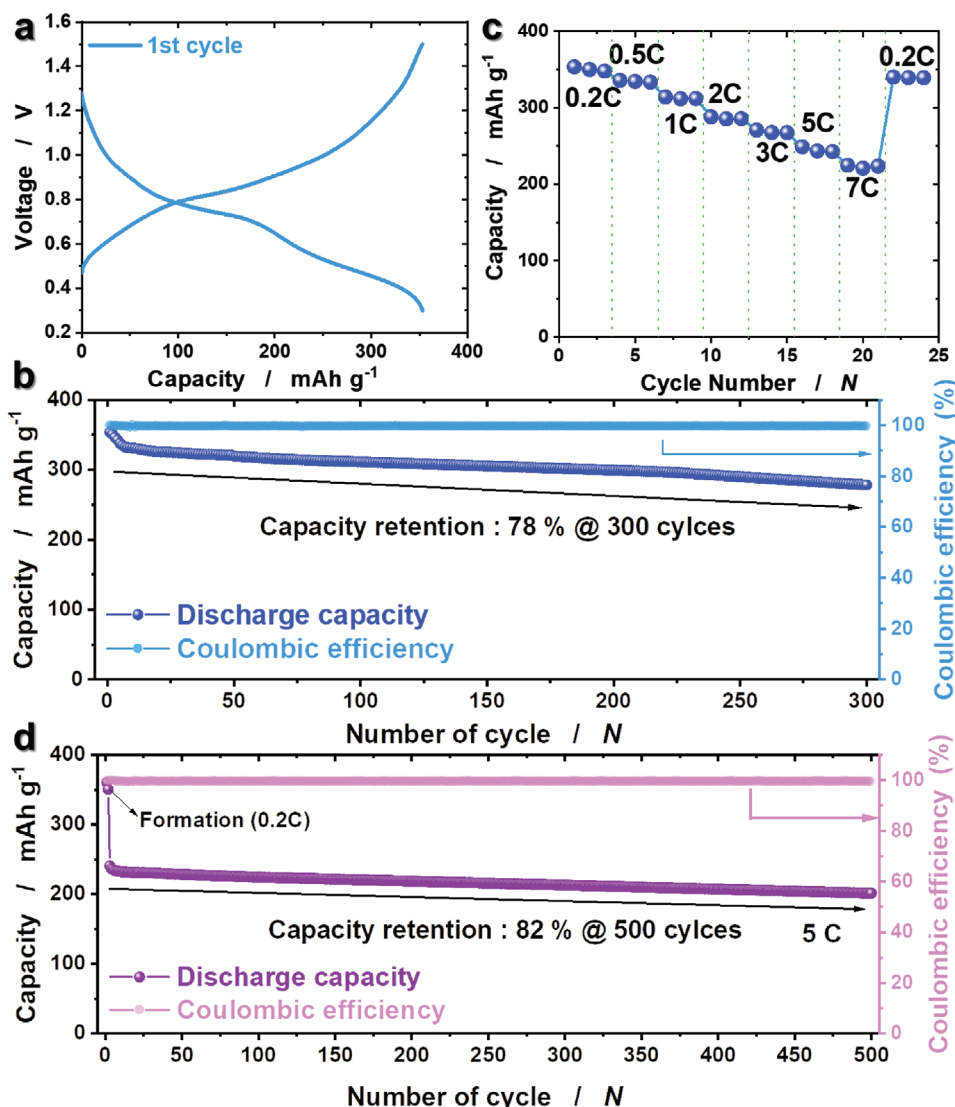


Figure 4. a) First discharge and charge curve of NaV₃O₈; b) corresponding cyclability at 0.2 C; c) rate capability at various rate of 0.2–7 C; d) high rate (5 C) cyclability for 500 cycles.

was tested at various currents from 0.2 C (70 mA g⁻¹) to 7 C (2.45 A g⁻¹) in Figure 4c and Figure S1b (Supporting Information). The NaV₃O₈ electrode was able to deliver a high capacity up to a rate of 7 C (2.45 A g⁻¹), with a high discharge capacity of 225 mAh g⁻¹. Excellent recovery capability was confirmed with the delivery of a discharge capacity of 340 mAh g⁻¹ at 0.2 C after the high-rate test at 7 C. In addition, less capacity was delivered below 0.6 V at high C-rates, which may be associated with complex kinetics in the voltage range. The NaV₃O₈ electrode delivered a high discharge capacity of 240 mAh g⁻¹ at a rate of 5 C (1.75 A g⁻¹) with acceptable capacity retention of ≈82% over 500 cycles (Figure 4d and Figure S1c, Supporting Information). To confirm the Na⁺ release from the NaV₃O₈ during cycling, the electrolytes were analyzed after 1st and 10th cycles by ICP-AES, indicating no trace of sodium element (Table S3, Supporting Information). That is, the sodium ions positioned between the vanadate layers played a role as a pillar between the vanadium oxide layers, which stabilizes the host structure during repeated

Zn²⁺ insertion and extraction. This pillar effect of alkali ions on structural stability is known for alkali-vanadate materials such as LiV₃O₈, Na_{1.1}V₃O_{7.9}, and K_xV₃O₈.^[32,35,37,42] Therefore, the open structure of NaV₃O₈ accommodates repeated expansion and contraction of the structure induced by de/intercalation of Zn²⁺ ions upon prolonged cycles, leading to the reasonable capacity retention and good rate capability with high capacities.

Moreover, we performed nudged elastic band (NEB) analyses based on first-principles calculation to determine the theoretical properties of NaV₃O₈ related to the power-capability. **Figure 5** shows the theoretically predicted Zn²⁺ diffusion motion in the Zn_xNaV₃O₈ structure and the activation barrier energy required for Zn²⁺ diffusion. It was verified that an activation energy of 423 meV is required for Zn²⁺ diffusion along the *bc* plane in the structure, which is a reasonable value that can enable facile Zn²⁺ diffusion in the structure. Thus, we confirmed that NaV₃O₈ can exhibit reasonable power-capability as a promising electrode for ZIBs.

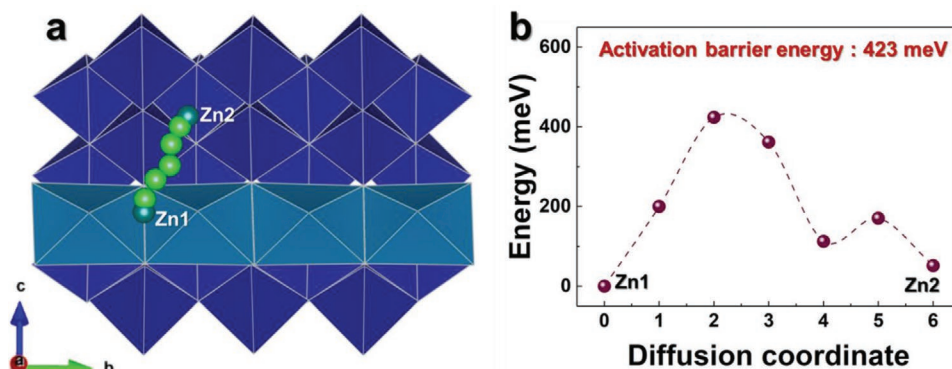
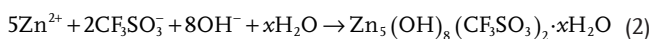
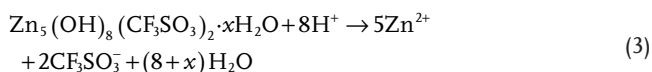


Figure 5. a) Predicted Zn^{2+} diffusion motion in ZnNaV_3O_8 structure and b) theoretically predicted activation barrier energies for Zn^{2+} diffusion in ZnNaV_3O_8 structure.

To rationalize the electrochemical reaction associated with Zn^{2+} intercalation based on above Figures 4 and 5, the NaV_3O_8 electrode was monitored using operando synchrotron XRD (o-SXRD) for two cycles (Figure 6a). The main peaks observed at 7.6° , 12.6° , 23° , 25.9° , 27.9° , 28.7° , 29.6° , and 38.8° (2θ), corresponding to the (001), (100), (003), $(-0-21)$, (110/11-1), (201/10-4/11-2), and (30-3) planes, respectively, shifted toward lower angles during discharge (0.3 V). Upon charging to 1.5 V, in contrast, the peaks reversibly returned to the original Bragg peak positions of NaV_3O_8 . In addition, the same tendency was observed during the second cycle. The calculated lattice parameters reveal a gradual increase in the b - and c -axes but decrease in the a -axis on discharge and during charge vice versa (Figure 6b). These results suggest that Zn^{2+} ions were de/intercalated into/from the open structure. In addition, new XRD peaks gradually appeared at 6.8° , 13.3° , and 32.9° upon reduction, whereas these peaks disappeared upon oxidation. These peaks may indicate formation of a new phase, that is formed during reduction. As we mentioned before, $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot 5\text{H}_2\text{O}$ was identified as the emerging phase from ZnSO_4 electrolyte. Thus, we assumed formation of a new Zn layered hydroxyl salt, which contains CF_3SO_3^- instead of sulfates, since electrolyte in our work contained triflate ions. According to Zn salts chemistry, Zn forms hydroxy salts with single-charged anions (An) of common formula $\text{Zn}_5(\text{OH})_8(\text{An})_2 \cdot x\text{H}_2\text{O}$. In particular, $\text{Zn}_5(\text{OH})_8(\text{CH}_3\text{COO})_2 \cdot x\text{H}_2\text{O}$ has very close XRD pattern to that of emerging phase in triflate electrolyte.^[43] Therefore, in our case the formed phase was found to be $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ having triflate ion in the structure. This compound deposits upon reduction as an interfacial layer and dissolves upon oxidation. The reason for $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ formation is related to local pH increase during discharge (reduction) process, which causes precipitation of the basic salt, as follows:



The reverse potential scan (oxidation) results in dissolution of precipitates:



We further examined the interfacial layer of the fully discharged and charged NaV_3O_8 electrodes using TEM-EDS and SEM-EDS mappings (Figure S2, Supporting Information, Figure 6c,d). As expected from o-SXRD, TEM observation of the NaV_3O_8 electrode discharged to 0.8 V showed stick-shaped (NaV_3O_8) and plate-like ($\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$) particles (Figure S2a, Supporting Information). Discharging to 0.3 V, new particles with flower-like shapes grew on the electrode surface, as observed in Figure 6c. The elemental mapping images revealed the uniform distribution of carbon (C), fluorine (F), sulfur (S), oxygen (O), and zinc (Zn) elements on the new particles for the discharged $\text{Zn}_2\text{NaV}_3\text{O}_8$ electrode; however, vanadium (V) element was not detected together with the new particles. TEM observation also confirmed the emergence of the large plate-like particle (Figure S2b, Supporting Information). TEM-EDS mappings, tilted 45° for observation of Figure S2b (Supporting Information), exhibited distribution of C, F, S, O, and Zn elements, which may be defined as $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ particle (Figure S2c, Supporting Information). On the contrary, the plate-like particles were not observed after charging to 0.8 V (Figure S2d, Supporting Information) and 1.5 V (Figure S2e, Supporting Information), and Zn, S, and F elements were not visible (Figure 6d). These results support the above o-SXRD findings that the interfacial $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ compound emerged during Zn^{2+} insertion and disappeared upon Zn^{2+} extraction from the host structure. Figure 6e presents a 2D map of the operando Raman spectra obtained in 1 M $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ aqueous solutions upon discharge and charge, with representative spectra shown in Figure S3 (Supporting Information). The group of peaks below 800 cm^{-1} corresponds to the signal from the NaV_3O_8 material with five main peaks located at 129, 285, 476, 549, and 762 cm^{-1} .^[44] The details of the spectra related to NaV_3O_8 are presented in Figure S4 (Supporting Information). In addition, anions of the electrolytic salt and conducting carbons were observed in the spectra, namely, at 318, 351, 577, 764, 1032, 1183, 1227 cm^{-1} for the vibrations of the triflate (CF_3SO_3^-) anion and at 1597 and 1350 cm^{-1} for carbons.^[45] An important feature is that the peaks belonging to NaV_3O_8 decreased during discharge (reduction), whereas the opposite effect was observed on charging (oxidation). In addition, the background signal increased when the signals of NaV_3O_8 were observed. It should

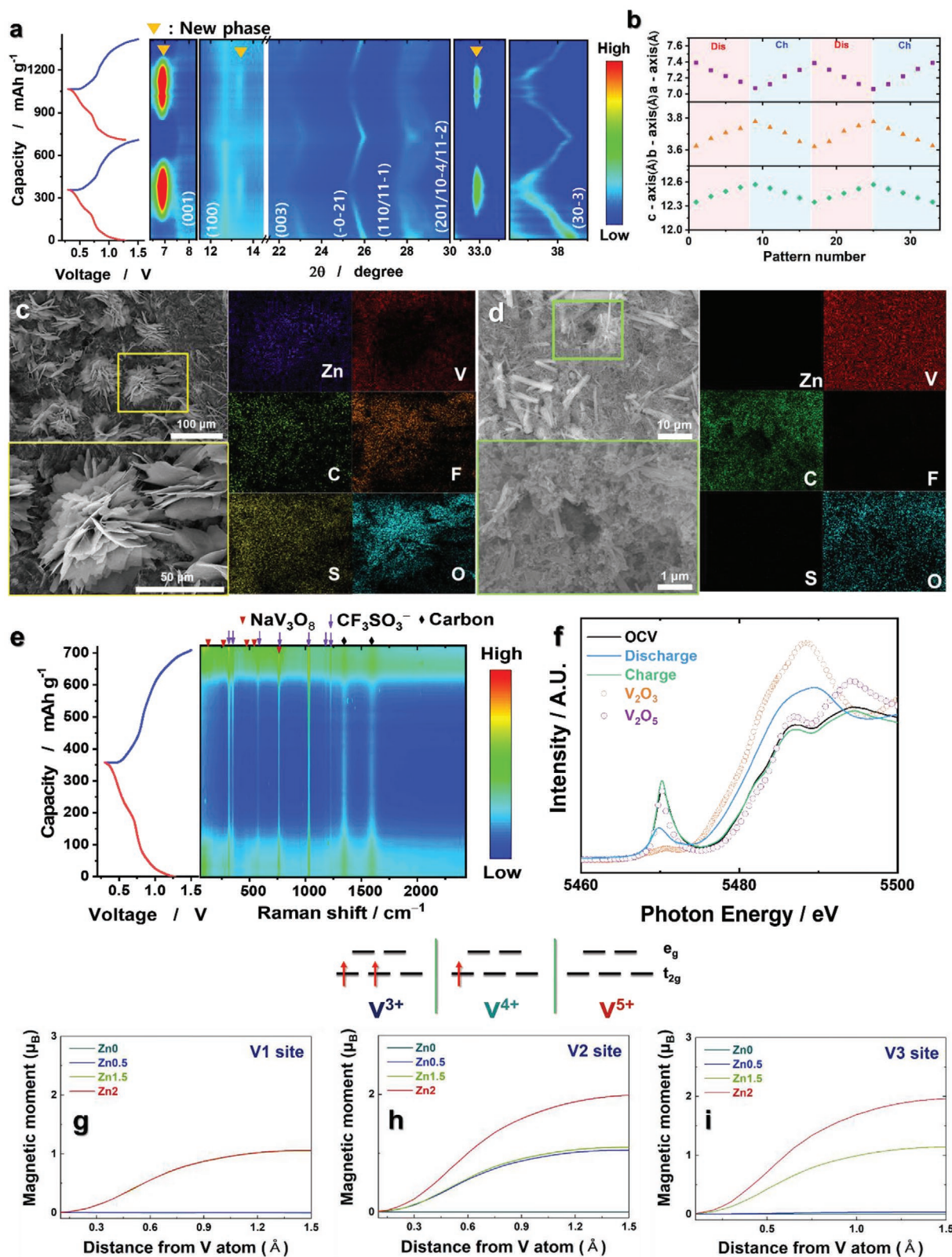


Figure 6. a) Operando-SXRD results of NaV₃O₈ during two cycles; b) the calculated lattice parameters obtained from o-SXRD data; SEM with EDS mapping images of c) fully discharged (0.3 V) electrode and d) fully charged (1.5 V) electrodes; e) 2D map of operando Raman spectra of NaV₃O₈-based cathode material in 1 m Zn(CF₃SO₃)₂ solutions; f) XANES result for V K-edge during initial cycle. The integrated spin moments of V ions on g) V1, h) V2, and i) V3 sites in various Zn_xNaV₃O₈ phases (x = 0, 0.5, 1.5, and 2).

be noted that the appearance of active material signals occurred over 0.8 V, whereas the signal disappeared below 0.8 V. This tendency is consistent with the *o*-SAXRD data (Figure 6a). Therefore, the disappearance of the NaV_3O_8 signal can be explained by the growth of the new phase, $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$, as the interfacial layer, as shown in Figure 6c. This new phase covers the surface preventing the light penetration during Raman signal acquisition. To identify the average oxidation state of vanadium during the electrochemical reaction of NaV_3O_8 in Zn cells, ex situ XANES measurement was performed for the fresh, fully discharged (0.3 V), and charged (1.5 V) electrodes (Figure 6f). The V K-edge spectrum of the fresh NaV_3O_8 shifted to the lower-energy region, and the relative intensity of the pre-edge was evidently reduced, indicating reduction toward V^{3+} , presumably 3.6+ according to the linear fit, from V^{5+} through the electrochemical reduction process. The V K-edge spectrum returned to a similar energy region as the fresh NaV_3O_8 in the fully charged state (1.5 V). The *o*-SAXRD, *o*-Raman spectroscopy, and ex situ XANES results verify that the open-structured NaV_3O_8 allows $\approx p$ mol of Zn^{2+} ions accompanied by $\text{V}^{5+/3+}$ redox pairs in the voltage range of 0.3–1.5 V, accompanied by the formation and extinction of $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ as the interfacial layer on the surface of NaV_3O_8 . The redox reaction of $\text{Zn}_x\text{NaV}_3\text{O}_8$ during Zn^{2+} de/intercalation was also confirmed through first-principles calculation. Figure 6g–i shows the integrated spin moments of V ions on each V site in the $\text{Zn}_x\text{NaV}_3\text{O}_8$ structure. The magnetic moments of the V1 and V2 sites changed from 0 to 1 Bohr magneton during the intercalation of 2 mol Zn^{2+} , and the V3 site experienced variation of the magnetic moments between 0 and 2. These results imply the $\text{V}^{3+}/\text{V}^{4+}/\text{V}^{5+}$ redox reaction of $\text{Zn}_0\text{NaV}_3\text{O}_8$ during charge/discharge, which is consistent with the ex situ XANES analyses on $\text{Zn}_2\text{NaV}_3\text{O}_8$. Furthermore, we compared the calculated lattice parameters by first-principle calculation and the values obtained by *o*-SAXRD (Table S4, Supporting Information). As predicted by the calculation, the unit volume expanded as Zn^{2+} insertion was progressed. The selected *o*-SAXRD patterns from Figure 6a also exhibited monotonous change in the XRD patterns (Figure S5, Supporting Information). In addition, the average oxidation state of V was reduced toward V^{3+} on discharge and oxidized to V^{5+} on charge (Figure 6f). Namely, as predicted, Zn^{2+} insertion induced expansion of the unit volume together with reduction of the average oxidation state of V toward 3+.

Zn^{2+} ions are transferred through the layered $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ to be intercalated into the NaV_3O_8 in the low-voltage region. Thus, further analyzed the electrode using cyclic voltammetry (CV) combined with electrochemical impedance spectroscopy (EIS) to understand the electrochemical behavior in the low-voltage region (Figure 7a–c). EIS measurements were performed at different potentials by recording and analyzing spectra at each 0.1 V step of the cyclic potential staircase (Figure 7a). Two distinct types of spectra were obtained depending on the electrode potential. Figure 7b presents a typical spectrum at relatively high electrode potentials (above 0.8 V), which consists of a charge-transfer arc in the high-frequency domain and a sloping part in the low-frequency domain. The equivalent circuit presented in Figure 7b is based on a Frumkin–Melik-Gaikazyán (FMG) circuit, which was shown to describe intercalation reactions in some systems.^[46,47] It

contains pseudo-capacitance, C_p , the circuit element originally attributed to reversible electrochemical adsorption in FMG model, but in case of battery material representing intercalation pseudocapacitance.^[43] Warburg impedance Z_{W1} results from diffusion of ions, and charge-transfer resistance R_{ct} is a kinetic characteristic of electrochemical reaction. Z_{W2} is Warburg impedance which results from diffusion-controlled charge transfer via an additional pathway. In contrast, when the electrode potential was fixed at relatively low values, typically more negative than 0.8 V, a second arc appeared in the Nyquist plot (Figure 7c). The latter arc corresponds to a lower-frequency domain than that of the first arc and may correspond to an additional charge-transfer process that emerges only at low potentials. The frequency response differs insignificantly in both directions of the potential variation. The equivalent circuit contains two consecutively connected blocks with two charge-transfer resistances (R_{ct} and R_{ct}'), which is often caused by the growth of a solid electrolyte interface (SEI) in batteries.^[48–50] Hence, it is thought that the additional charge-transfer resistance below 0.8 V corresponds to Zn^{2+} transfer through an interfacial layer to the cathode material, whereas the high-frequency charge-transfer resistance represents the charge-transfer resistance at the solid/liquid interface; that is, it corresponds to the electrolyte/interfacial layer instead of the electrolyte/ NaV_3O_8 interface. Moreover, the variation in the EIS spectra with potential is especially distinct in the phase shift part of the Bode plot (Figure S6a,b, Supporting Information). The peak at ≈ 300 –600 Hz corresponds to a small arc in the Nyquist plots, which results from the charge transfer observable in the entire potential range. The peak at 1–10 Hz, in contrast, appears only at relatively low potential below 0.8 V and vanishes above 0.9 V in the reverse potential variation. Because of the drastic change in the impedance spectra at ≈ 0.8 V, different equivalent circuits were observed to fit the experimental data in different potential ranges (Figure 7d). Figure 7e shows the charge-transfer resistance (R_{ct}) at the electrode/liquid electrolyte interface, whereas Figure 7f shows the additional charge-transfer resistance R_{ct}' which arises in series with R_{ct} below 0.8 V, due to formation of the interfacial layer and the corresponding interface between the interfacial layer and cathode material. R_{ct} increased slightly with the extent of discharge and drastically fell at 0.8 V. At the same time, R_{ct}' emerged and increased with the extent of discharge with a local minimum at 0.6 V. The huge increase from 0.8 to 0.4 V of R_{ct}' comes from growth of the interfacial layer. A schematic representation of the described process and related parameters is summarized in Figure 7g. The hysteresis of the parameter variation in the reverse change of the potential is evidently due to the interfacial layer formation below 0.8 V and its disappearance above 0.8 V. As seen from the SEM-EDS (Figure 6c,d) and TEM-EDS (Figure S2, Supporting Information) results, it is most likely that the formation of $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ interfacial layer is preferentially stimulated on the most electroactive sites from 0.8 V on discharge. Combined with Figures 6 and 7, therefore, it could conclude that Zn^{2+} ions are transferred through the layered $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ layers to be intercalated into the NaV_3O_8 .

Furthermore, we have studied similar phenomena with ex situ XRD (Figure S7a, Supporting Information), SEM (Figure S7b, Supporting Information), EIS (Figure S8,

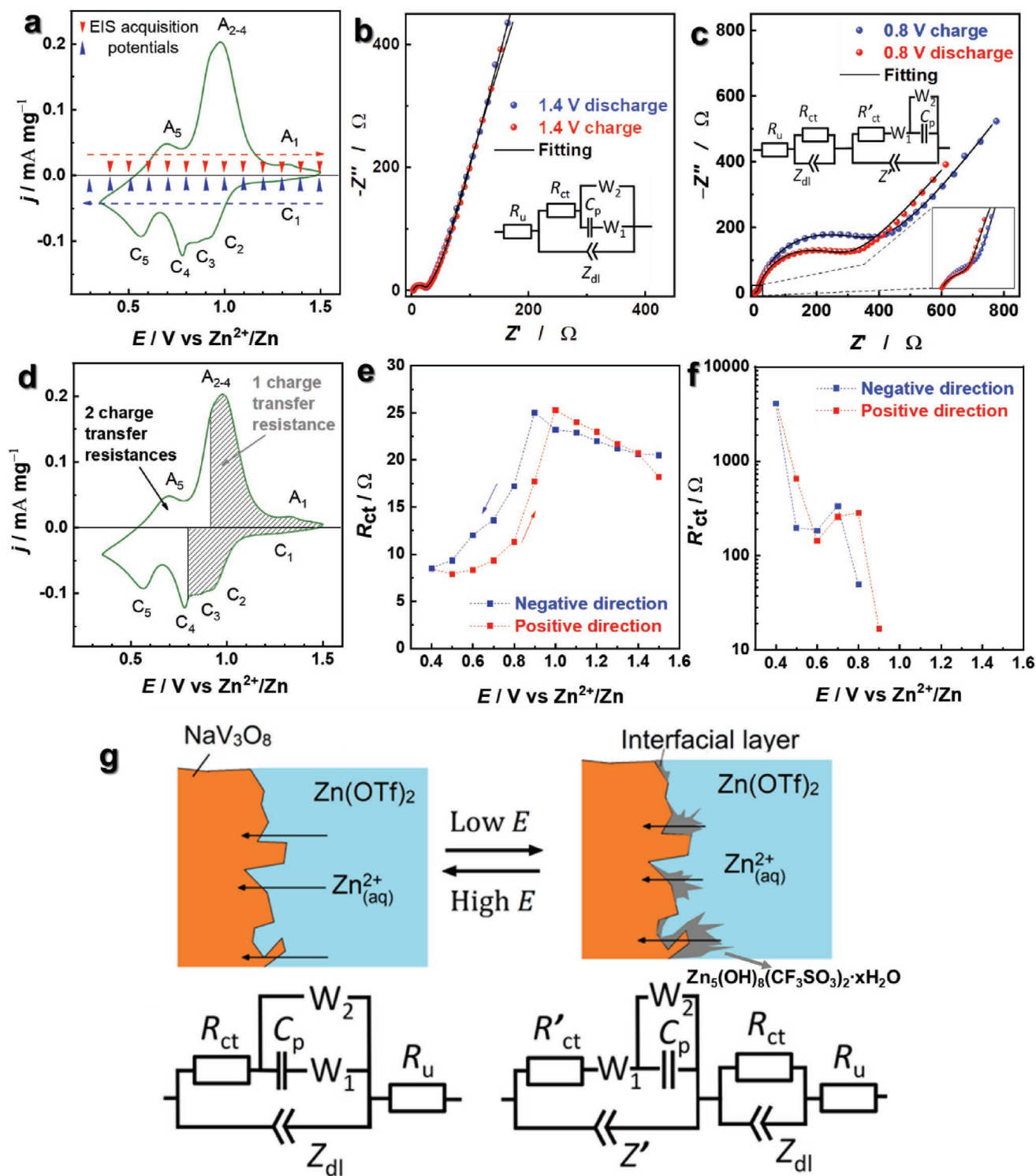


Figure 7. EIS characterization of NaV_3O_8 . a) Potentials of spectra acquisition versus CV. Experimental and fitted spectra at b) 1.4 V and c) 0.8 V with the corresponding equivalent electrical circuits; d) CV with highlighted regions showing similar type of frequency response; dependence of e) interfacial charge-transfer resistance (R_{ct}) and f) charge transfer from SEI (R'_{ct}) on potential; g) schematic representation of hypothetical SEI formation at low potentials and corresponding changes in equivalent electrical circuit.

Supporting Information), and operando Raman spectra (Figure S9, Supporting Information) in ZnSO_4 electrolyte. Ex situ XRD data and SEM images (Figure S7, Supporting

Information) presented formation of a new product, $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$, after discharging to 0.3 V, which is similar to the formation of $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ in $\text{Zn}(\text{CF}_3\text{SO}_3)_2$

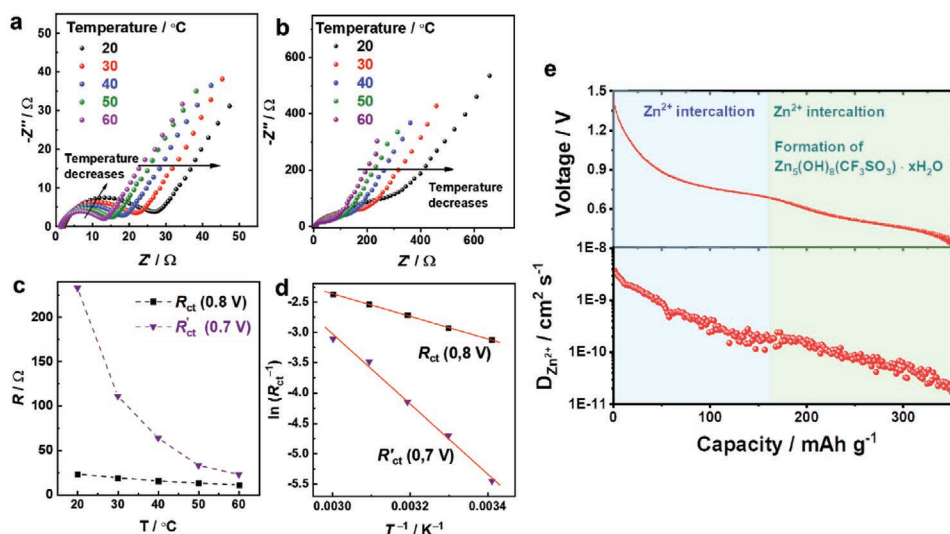


Figure 8. Impedance spectra of NaV_3O_8 electrode at different temperatures at a) 0.8 V and b) 0.7 V; c) R_{ct} and R_{ct}' dependences on temperature derived from the spectra shown above; d) Arrhenius plot for charge-transfer resistances R_{ct} and R_{ct}' ; e) GITT results and Zn^{2+} diffusivity measured by GITT for NaV_3O_8 electrode.

electrolyte (Figure 6). According to literature by Soundharrajan et al.,^[36] $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$ was produced and disappeared upon reduction/oxidation reaction. For EIS spectra (Figure S8, Supporting Information), a new charge-transfer resistance appeared around 0.9 V upon discharge and increased as potential lowered. Operando Raman spectra of NaV_3O_8 in ZnSO_4 aqueous solution also exhibited the similar tendency compared to phenomena seen in $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte, indicating that $\text{Zn}_4(\text{OH})_6\text{SO}_4 \cdot x\text{H}_2\text{O}$ was generated on discharge and extinguished on charge (Figure S9, Supporting Information).

As evident from the above results in Figures 6 and 7, the charge-transfer processes represented by the charge-transfer resistances R_{ct} (above 0.8 V governed by Zn^{2+} intercalation) and R_{ct}' (below 0.8 V progressed with Zn^{2+} intercalation and the formation of $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$) should have different activation energies. The latter can be derived from the corresponding dependence of the charge-transfer resistance on temperature because the charge-transfer resistance is reciprocal to the rate constant (k):

$$R_{\text{ct}} \sim k^{-1} \quad (4)$$

Considering the Arrhenius equation, the natural logarithm of the reciprocal charge-transfer resistance is proportional to the reciprocal absolute temperature:

$$\ln(R_{\text{ct}}^{-1}) = \text{constant} - \frac{E_a}{RT} \quad (5)$$

where E_a is the activation energy, R is the universal gas constant, and T is the absolute temperature. This approach has been successfully applied in the investigation of several battery materials.^[48,51] Using a similar approach, we analyzed the EIS spectra at different temperatures at 0.8 and 0.7 V (a drastic change in the system occurs in this potential region). Figure 8a,b shows the variation of the impedance spectra with temperature at

0.8 and 0.7 V, respectively. The spectra fitting was performed using the equivalent electrical circuits shown in Figure 7b,c. The resulting dependences of R_{ct} and R_{ct}' on temperature are given in Figure 8c, and the resulting Arrhenius plots are presented in Figure 8d. According to Equation (5), the slope of the latter linear dependences is equal to $-E_a/R$. Thus, the values of E_a 0.8 V (R_{ct}) and E_a 0.7 V (R_{ct}') were calculated to be 15.5 ± 0.2 and 48 ± 2 kJ mol⁻¹, respectively. Thus, the emerging charge-transfer process (R_{ct}') has a higher activation energy than that of the charge transfer (R_{ct}) on the solid/liquid interface, which slows down the charge transfer and may limit the power-capability below 0.8 V, as Zn^{2+} ions intercalate through the $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ interfacial layer in the low-voltage range to reach the cathode material, which is associated with an additional charge-transfer step with high activation energy. Galvanostatic intermittent titration technique (GITT) measurements were performed to calculate the diffusion coefficient ($D_{\text{Zn}^{2+}}$) to compare the diffusivity of Zn^{2+} for NaV_3O_8 for the two plateaus of 0.7–1.5 and 0.3–0.7 V (Figure 8e). During the discharge process, the measured diffusion coefficient of Zn^{2+} ions for the first plateau ranged from 10^{-8} to 10^{-10} cm² s⁻¹; in addition, a lower diffusion coefficient of 10^{-10} to 10^{-11} cm² s⁻¹ was observed for the second plateau. It is apparent that the fade of the diffusion coefficient for the second plateau is mainly affected by the formation of the $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ interfacial layer on NaV_3O_8 . This finding coincides with the electrode performances shown in Figure 4, and the formation of $\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$ results in the deterioration of facile de/intercalation of Zn^{2+} into/from the open-structured NaV_3O_8 . To improve electrode performance based on this finding, we need to explore use of highly concentrated electrolyte. Since the $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ salt has bulky anions, the highly concentrated $\text{Zn}(\text{CF}_3\text{SO}_3)_2$ electrolyte may reduce solvation effect.^[52–55] Another strategy is to use water-in-salt electrolyte, which was shown to improve capacity retention in ZIBs.^[56] Therefore, it is anticipated to improve Zn^{2+} ion transfer and long-term cyclability in Zn cells.

3. Conclusion

In summary, through combined studies using various experimental techniques and first-principle calculations, we confirm the theoretical electrochemical properties and reaction mechanism of NaV_3O_8 during Zn^{2+} de/intercalation. This material delivers capacities of 353 mAh g^{-1} at 0.2 C and 240 mAh g^{-1} at 5 C , with capacity retention of 82% after 500 cycles. The insertion of Zn^{2+} changes the oxidation state of vanadium from V^{5+} to V^{3+} , allowing 2 mol of Zn per formula unit. Electrochemical study reveals an abrupt change of response at $\approx 0.8 \text{ V}$ versus Zn^{2+}/Zn , with the emergence of a new charge-transfer resistance at low potentials. The EIS-determined charge-transfer activation energy above 0.8 V is observed to be 16 kJ mol^{-1} , whereas an additional charge-transfer step below 0.8 V results in an activation energy of 48 kJ mol^{-1} and slows down the charge-transfer rate by a magnitude. Operando synchrotron XRD and Raman spectra analyses verified that the response change is caused by the formation of a zinc hydroxytriflate ($\text{Zn}_5(\text{OH})_8(\text{CF}_3\text{SO}_3)_2 \cdot x\text{H}_2\text{O}$) interfacial layer below 0.8 V , which disappears above this voltage. The interfacial zinc hydroxy salt formation appears to be a common feature for many cathode materials in aqueous media during the electrochemical process. Understanding of the interfacial layer formation or new phase precipitation at cathodes is of great importance for ZIB design because these processes may reduce the charge-transfer rate and may be a limiting factor in the creation of high-power-density ZIBs.

4. Experimental Section

NaV_3O_8 was synthesized through a hydrothermal process using NaVO_3 (Sigma-Aldrich) and 0.1 M HCl (Sigma-Aldrich). First, NaVO_3 was dissolved in deionized (DI) water and stirred for 30 min to prepare a homogeneous solution. Then, 0.1 M HCl solution was added to the NaVO_3 solution, and the resulting solution was stirred with a magnetic stirrer, resulting in a gradual change in the solution color from orange to brown. Finally, the brown-colored solution was transferred into a Teflon-lined autoclave and heated at 180°C for 12 h. After the reaction, the produced powders were washed with DI water several times and dried at 70°C for 1 day. The obtained materials were grinded and heated at 400°C for 4 h in dry air.

The crystal structure of the product was identified using powder X-ray diffraction (XRD, PANalytical, Empyrean) with $\text{Cu K}\alpha$ radiation; the measurements were performed in the range of $10^\circ \leq 2\theta \leq 80^\circ$ (2θ) with a step size of 0.03° (2θ). The obtained XRD patterns were analyzed using the FullProf Rietveld program. Field-emission scanning electron microscopy (FE-SEM, JEOL, JXA-8100) and high-resolution transmission electron microscopy (HR-TEM, JEOL, JEM-3010) were employed to observe the particle morphology. X-ray photoelectron spectroscopy (PHI 5600, PerkinElmer) was used to observe the chemical state and surface state of the NaV_3O_8 electrode in macro mode ($3 \times 3 \text{ mm}^2$) using a Mg X-ray source. The chemical composition of the product was quantified using inductively coupled plasma-atomic emission spectroscopy (ICP-AES; OPTIMA 8300, Perkin-Elmer).

The produced active material was composited with electro-conducting agents (Super P and Denka black, 1:1 in wt%) and a binder (carboxymethyl cellulose, CMC) at a ratio of 8:1:1 in DI water. The slurry was applied onto stainless-steel mesh that was used as the current collector. R2032 coin-type cells were paired with the cathode and Zn metal as the anode, with the electrodes separated by a glass filter as the separator in the presence of $1 \text{ M Zn}(\text{CF}_3\text{SO}_3)_2$ (zinc trifluoromethanesulfonate or zinc triflate) aqueous solution as the

electrolyte. The cells were first subjected to discharge (reduction) and then charge (oxidation) in the voltage range of $0.3\text{--}1.5 \text{ V}$ at a rate of 0.2 C (70 mA g^{-1}) for long cycling at various rates from 0.2 to 7 C at 25°C . Galvanostatic intermittent titration technique (GITT) measurements were performed to calculate the diffusion coefficient ($D_{\text{Zn}^{2+}}$) Zn^{2+} for NaV_3O_8 . The electrode was measured at each 10 min step of discharge (Zn^{2+} intercalation)/charge (Zn^{2+} deintercalation) followed by a 30 min relaxation step in a Zn cell.

CV, EIS, and operando Raman were collected in a three-electrode cell with Zn counter and Zn reference electrodes in $1 \text{ M Zn}(\text{CF}_3\text{SO}_3)_2$. Metrohm Autolab was used to collect three-electrode cyclic voltammetry (CV) data, electrochemical impedance spectroscopy (EIS) measurements (including activation energy), and operando Raman spectra. EIS spectra were collected each 0.1 V using a 10-mV amplitude in the frequency range of $5 \text{ mHz--}100 \text{ kHz}$. The conditioning time was 1 h before each spectrum was collected, and the obtained EIS data were analyzed using EISSA software.^[57]

We performed first-principle calculations based on density functional theory (DFT) calculations using the Vienna Ab initio Simulation Package (VASP).^[58] The projector-augmented wave (PAW) pseudopotential^[59] and Perdew–Burke–Ernzerhof (PBE) parametrization of the generalized gradient approximation (GGA)^[60] were applied for the DFT calculations. For the localization of the d -orbital in the V ions,^[61] the GGA+U method was adopted with a U value of 4.2 eV , which was previously used for DFT calculation on V-based electrode materials for ZIBs.^[62,63] All the calculations were performed with an energy cutoff of 500 eV until the remaining force in the system converged to less than $0.03 \text{ eV \AA}^{-1}$ per unit cell. The cluster-assisted statistical mechanics (CASM) software^[64] was used for preparation of various Zn^{2+} /vacancy configurations for each $\text{Zn}_x\text{NaV}_3\text{O}_8$ composition; this step was followed by full DFT calculations for a maximum of 30 configurations with the lowest electrostatic energy for each composition to obtain a convex-hull plot for $\text{Zn}_x\text{NaV}_3\text{O}_8$. For the nudged elastic band (NEB) method,^[65] five intermediate images were prepared through linear interpolation of the initial and final images after the initial and final $\text{Zn}_x\text{NaV}_3\text{O}_8$ configurations were determined. For calculation of activation barrier energies using the NEB method, the lattice parameters of the $\text{Zn}_x\text{NaV}_3\text{O}_8$ configurations were fixed, and all the internal degrees of freedom were allowed to relax.

To identify the structural transition, operando synchrotron XRD (o -SXRD), and ex situ X-ray absorption near edge structure (XANES) spectroscopy were performed during the discharge (reduction) and charge (oxidation) processes. o -SXRD analysis was conducted at the 3D X-ray scattering (XRS) beamline at the Pohang Accelerator Laboratory (PAL), Pohang, South Korea, and the cell was measured in the 2θ range between 10° and 60° . Operando Raman spectra were recorded using a Nanofinder HE (LOTIS TII, Belarus HE) test confocal spectrometer (Figure S10, Supporting Information). For the XANES measurement, the cells were carefully disassembled, and the electrodes were washed with DI water. Then, the electrodes were dried at 80°C in a vacuum oven overnight. The XANES measurements were performed at the 7D X-ray absorption fine structure (XAFS) beamline at the Pohang Accelerator Laboratory (PAL), Pohang, South Korea. The K-edge XANES data were collected in total electron yield mode, and the sample current was recorded. Ion chamber detectors filled with high-purity N_2 gas were used to record the X-ray absorption spectra.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

batteries, intercalation, operando, vanadate, zinc

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